Thematic Analysis of C-MOOR Research Posters

Research Motivation Categorization

Minimax M2 AI

April 2026

Executive Summary

This report presents a thematic analysis of 27 research posters from the C-MOOR genomic data science training program. The analysis categorizes posters based on the primary motivation behind the research, focusing on whether the work is driven by understanding fundamental biological mechanisms or by investigating human disease processes.

Total Posters Analyzed: 27

Part I: Category Summary Table

Category	Count	Posters
Basic biology (Neurology)	3	Gill & Alcazar (2025); Godoy-Pena et al. (2025); Luera et al. (2025)
Basic biology (Metabolism)	6	Berbouti et al. (2025); Camara et al. (2025); Dehuelbes et al. (2025); Lemus et al. (2025); Nii et al. (2025); Pedireddi et al. (2025)
Basic biology (Other)	8	Diaz et al. (2025); Ford et al. (2023); Henriquez et al. (2025); Holmes (2025); Meraz et al. (2023); Sakana et al. (2025); Tuttle et al. (2025); Zlaket et al. (2023)
Disease research (Neurology)	3	Avetisyan et al. (2023); Cabral et al. (2025); Haubelt & Alcazar et al. (2025)
Disease research (Metabolism)	5	Brown et al. (2025); Logan et al. (2025); Otala et al. (2025); Rodriguez (2025); Trevino et al. (2022)

Category	Count	Posters
Disease research (Other)	2	Paderna et al. (2025); Paramo-Ojeda et al. (2025)

Part II: Detailed Analysis with Supporting Evidence

1. Basic biology (Neurology)

Gill & Alcazar (2025) - CCC FA25.txt

Title: Differential Expression of Serotonin Receptor Genes Htr1a and Htr2a in the Prefrontal Cortex and Striatum

Evidence for this categorization: - Primary focus: Serotonin receptor gene expression in different brain regions - Introduction: 'the role of the prefrontal cortex is decision making and emotional regulation' and 'the striatum role is mainly motor coordination and reward processing' - Research question: How are serotonin receptors differentially expressed in these brain regions? - No named neurological disease is the primary motivation - Focus is on understanding fundamental neurotransmission patterns

---

Godoy-Pena et al. (2025) - CCC FA25.txt

Title: Regulation of Sleep Quality: Analysis of Gene Expression Using Bulk RNA-seq Data

Evidence for this categorization: - Primary focus: Sleep regulation and the gut-brain connection - Abstract: 'determine what genes dictate the kind of sleep drosophila flies get, and connect it to a gut-brain region' - Research on CCHa1-R, CCHa2-R, NPF - genes involved in sleep-wake behaviors - Introduction: 'our goal was to identify similarities in these expression patterns to gain a better understanding of how these genes may influence sleep quality' - Sleep is a fundamental biological process, not a named disease. While sleep disorders are mentioned in background, the research focuses on understanding normal sleep regulation mechanisms

---

Luera et al. (2025) - CCC FA25.txt

Title: Region-Specific NPF/NPFR Signaling in the Drosophila Midgut

Evidence for this categorization: - Primary focus: Neuropeptide F (NPF) and its receptor NPFR in feeding behavior - Abstract: 'We analyzed Neuropeptide F (NPF) and its receptor NPFR in Drosophila melanogaster to explore their role in gut-mediated feeding behavior' - Introduction: 'Neuropeptide F is a conserved neuropeptide that regulates feeding, stress, and sleep behaviors' - Research focuses on understanding fundamental neuropeptide signaling mechanisms - No named disease is the primary motivation - the focus is on understanding how feeding behavior is regulated at a molecular level

---

2. Basic biology (Metabolism)

Berbouti et al. (2025) - CCC FA25.txt

Title: Trehalase Expression in the Drosophila Midgut: Differential Expression across Anterior and Posterior regions

Evidence for this categorization: - Primary focus: Trehalose catabolism pathway - how sugar is broken down and transported in the midgut - Abstract states: 'This study provides insight into the regulation of carbohydrate metabolism in flies and parallel sugar processing in humans' - Research question asks whether TREH, TRET1, and GLUT1 show similar expression patterns - a fundamental metabolism question - No named disease is mentioned as the main motivating factor - Introduction states Drosophila is used because 'its carbohydrate metabolism is similar to other organisms, including humans' - fundamental biology comparison

---

Camara et al. (2025) - LU SP25.txt

Title: Functional Difference Between Amy-D, Amy-P and Amyrel in Drosophila Midgut

Evidence for this categorization: - Primary focus: Understanding functional differences between amylase genes in digestion - Abstract: 'We hypothesized differential expression of these genes across the midgut' - Methods focus on comparing expression patterns between regions - Discussion: 'The Drosophila midgut tightly regulates amylase expression by gut region' - While diabetes and Alzheimer's are mentioned as possible connections, the primary research question is about fundamental enzyme function and regional specialization in digestion, not disease mechanisms

---

Dehuelbes et al. (2025) - LU SP5.txt

Title: Regional Expression of Key Metabolic genes in Drosophila Midgut: CRAT, MANF, NPY

Evidence for this categorization: - Primary focus: CRAT (carnitine O-acetyltransferase), MANF, NPY - metabolic genes involved in energy homeostasis - Introduction: 'Acetyl-CoA links carbohydrate, fat, & protein metabolism' and 'Regulates acetyl-CoA levels and mitochondrial energy balance' - Introduction: 'MANF Promotes neuron survival, regulates ER stress response' and 'NPY Controls appetite, energy balance, circadian rhythms' - Research question focuses on fundamental metabolic gene function - While future therapeutic applications are mentioned, the primary motivation is understanding metabolic gene function, not treating a named disease

---

Lemus et al. (2025) - FA25 CCC.txt

Title: Macromolecule Breakdown and Absorption: Analysis of Differential Expression Using Single Cell and Bulk RNA-seq Data

Evidence for this categorization: - Primary focus: Understanding how amylase genes (Amy-P, Amyrel) function in macromolecule breakdown and absorption - Abstract: 'Productive macromolecule breakdown relies on certain genes expressed in specific regions of the Drosophila midgut' - Research question: How are amylase genes expressed across different midgut regions? - Discussion: 'These findings reveal functional diversity within the amylase-family' and highlight 'compartmentalized process' - No named disease is the primary motivation - focused on understanding fundamental digestive processes

---

Nii et al. (2025) - CCC FA25.txt

Title: Iron-Induced Modulation of Acetyl-CoA Carboxylase (ACC) in Drosophila Melanogaster: Impacts on Lipid Metabolism and Energy Storage

Evidence for this categorization: - Primary focus: How iron exposure affects ACC expression and lipid metabolism - Abstract: 'We hypothesize that excess iron activates acetyl-CoA carboxylase, increasing malonyl-CoA production and driving fatty acid synthesis' - Introduction: 'ACC generates malonyl-CoA, a critical substrate for fatty acid elongation and a metabolic signal' - Research question: How does elevated iron influence ACC expression and lipid storage? - While metabolic disorders are mentioned in the abstract as potential future relevance ('providing insight that can extend to human metabolic diseases such as obesity, fatty liver disease'), the primary research focus is on understanding fundamental metabolic regulatory mechanisms

---

Pedireddi et al. (2025) - CCC SP25.txt

Title: Beyond the Anterior: Amylase Gene Expression Suggests Widespread Polysaccharide Digestion in the Drosophila Midgut

Evidence for this categorization: - Primary focus: Understanding sugar metabolism and polysaccharide digestion in the midgut - Abstract: 'We were interested in genes involved in sugar metabolism since Drosophila melanogaster lives on a high sugar diet' - Research question: Where are starch and sucrose metabolism genes expressed? - Discussion: 'Our findings suggest that not only carbohydrate breakdown occurs in the anterior region, but the posterior region as well' - No named disease is the primary motivation - focused on understanding fundamental digestive processes

---

3. Basic biology (Other)

Diaz et al. (2025) - CCC SP25.txt

Title: V-ATPase Gene Expression Reveals Cellular Defense Zones in the Copper Region of the Drosophila Viidgut

Evidence for this categorization: - Primary focus: V-ATPase genes and their role in phagosome acidification and cellular defense - Abstract: 'we examined the regional expression of V-ATPase genes involved in phagosome acidification' - Introduction: 'Phagosomes are part of the cell's internal defense and cleanup system' - fundamental cell biology - Research is about understanding cellular defense mechanisms, not a named human disease - Discussion mentions analogy to human stomach function but not a specific disease

---

Ford et al. (2023) - CCC SP23.txt

Title: Cytochrome P450 genes providing insecticide resistance found in al, p1 and p2_4 midgut regions of Drosophila

Evidence for this categorization: - Primary focus: Cytochrome P450 gene family and their role in insecticide resistance - Abstract: ‘Some Cytochrome P450 genes play an important role in the development of insecticide resistance’ - Research question: understanding which midgut regions show high coexpression of these genes - No named human disease is the main motivating factor - Uses Drosophila as a model for human gut function, but the focus is on fundamental detoxification mechanisms

---

Henriquez et al. (2025) - COD WI24.txt

Title: 6 Immune Pathways Exhibit Similar Patterns of Gene Expressions in Drosophila melanogaster Midgut

Evidence for this categorization: - Primary focus: Understanding how six immune pathways (Toll, JAK/STAT, IMD, etc.) are expressed across midgut regions - Abstract: ‘Measurement of gene expression in six immune pathways suggests similar levels of immune pathway protein expression’ - Introduction: ‘Understanding fundamental biological mechanisms’ through immune pathway analysis - While Crohn’s disease and IBD are mentioned as context for why this research might be relevant to human health, the primary research question focuses on understanding fundamental immune mechanisms in the gut - The research is about understanding how the immune system functions in different gut regions

---

Holmes (2025) - CCC SP25.txt

Title: Melanogenesis within the Drosophila Midgut: Neuromelanin and Dopamine

Evidence for this categorization: - Primary focus: Yellow gene function and neuromelanin production in the midgut - Abstract: ‘This research was focused in genes connected to melanogenesis (synthesis of melanin) and its antioxidant uses’ - Research question: Understanding why the yellow gene is highly expressed in the Fe region and its connection to neuromelanin - Parkinson’s is mentioned in the discussion as potential future relevance, but the primary research is about understanding fundamental melanin synthesis mechanisms - Uses model organisms to explore how genes function in normal biological processes

---

Meraz et al. (2023) - CCC SP23.txt

Title: Tyrosine Kinase Expression in the Midgut Suggests a Role in the Copper Region

Evidence for this categorization: - Primary focus: Src42A gene (tyrosine kinase family) and its expression in the midgut - Abstract: ‘Tyrosine kinases are enzymes related to cell growth and development’ - Introduction: ‘genes relating to the midgut have been discovered to be differentially expressed among its various subregions’ - While FRK (human ortholog) is mentioned as a tumor suppressor, the research question is fundamentally about understanding gene expression patterns across midgut regions, not about cancer mechanisms - Hypothesis tests whether tumor suppressor genes are equally expressed - fundamental cell biology question

---

Sakana et al. (2025) - CCC SP25.txt

Title: Differential Expression of the TALE Homeobox Genes in Drosophila Midgut patterning

Evidence for this categorization: - Primary focus: TALE homeobox genes and their role in midgut developmental patterning - Abstract: ‘This project analyzed and explored the expression patterns of the 7 TALE genes in the Drosophila midgut, and how they might work together’ - Introduction: ‘Gene expression patterns that control development and differentiation are often orchestrated by transcription factors’ - Research question: How do TALE genes control patterning in the midgut? - No named disease - focused on understanding fundamental developmental biology - Uses model organisms to explore how genes function in normal developmental processes

---

Tuttle et al. (2025) - CCC SP25.txt

Title: Gene Expression Patterns in the Tryptophan-Kynurenine Pathway in Drosophila Melanogaster

Evidence for this categorization: - Primary focus: Tryptophan metabolism and the kynurenine pathway - Abstract: ‘This study investigates the differential expression of genes regulating Tryptophan and its metabolism pathways’ - Introduction: ‘tryptophan is metabolized through the kynurenine pathway, this process has been linked to aging’ - Research question: How are tryptophan metabolism genes expressed across midgut regions? - Focus is on understanding fundamental metabolic pathways and their role in aging processes - While the research references connections to aging and lifespan, it does not focus on a named human disease as the primary motivation

---

Zlaket et al. (2023) - CCC SP23.txt

Title: ScpX: A Peroxisomal Gene and its Distinctive Paralogs

Evidence for this categorization: - Primary focus: ScpX gene and its paralogs - understanding peroxisomal gene function - Abstract: ‘we researched the ScpX gene and its paralogs, which are homologous genes arisen from duplication events in the genome’ - Introduction: ‘In humans, the SCP2 gene is bifunctional… Errors within this gene can cause Zellweger’s Syndrome’ - Research question focuses on understanding fundamental gene function and paralog divergence - While Zellweger syndrome is mentioned as context, the primary research question is about understanding how these genes function normally and how their paralogs differ - Discussion: ‘Further research could be done on the metabolic pathways the genes are involved in within flies to explain why they differ in function’

---

4. Disease research (Neurology)

Avetisyan et al. (2023) - CCC SP23.txt

Title: Expression Analysis of Autism Related Genes Across the D. Melanogaster Midgut

Evidence for this categorization: - Primary focus: ASD (Autism Spectrum Disorder)-related genes in the gut-brain axis - Introduction explicitly states: 'When communication between the midgut and the brain is disrupted, there is a higher probability of autism in an individual organism' - References multiple autism-related genes: SATB1, KCNQ3, ALDH5AT1, NCKAP1, TBL1XR1, HNRNPH2 - Discussion mentions: 'further research may show an increase in ASD when these gene interactions are interrupted' - The research is explicitly motivated by understanding the molecular basis of autism, a neuropsychiatric condition

---

Cabral et al. (2025) - CCC FA25.txt

Title: MECP2 Expression in Mouse Brain Regions

Evidence for this categorization: - Primary focus: MECP2 gene, linked to Rett syndrome - 'Mutations in MECP2 are linked to Rett syndrome and related neurodevelopmental disorders' - Introduction states: 'MECP2 dysfunction is known to cause Rett syndrome and contribute to a range of neurological impairments' - Abstract: 'insight into how dysregulation of this gene may contribute to neurological dysfunction' - Research explicitly motivated by understanding the molecular basis of Rett syndrome - References discuss MECP2 mutations and Rett syndrome extensively

---

Haubelt & Alcazar et al. (2025) - CCC FA25.txt

Title: Exploring DRD2: The Association of PD Risk with Dopamine Receptors in the Dopaminergic Synapse

Evidence for this categorization: - Primary focus: Parkinson's Disease - 'I was interested in the genes, pathways, and mechanisms related to the neuropsychiatric disorder Parkinson's Disease' - Abstract: 'Abnormal variants of the Drd2 gene has been implicated in many neurological disorders including Parkinson's Disease' - Introduction: 'Many neuropsychiatric disorders including Parkinson's Disease, Schizophrenia, and Post-traumatic Stress Disorder are seen to have direct correlation to dopamine receptor D2' - Discussion: 'Observing differential expression… will illustrate… their role in the relationship between the dopaminergic synapse and neuropsychiatric disorders' - Parkinson's Disease is explicitly the main motivating factor

---

5. Disease research (Metabolism)

Brown et al. (2025) - CCC SP25.txt

Title: Analysis Of Gene Expression Of Acids Reflux Related Genes Across The Midgut

Evidence for this categorization: - Primary focus: Acid reflux disease - 'Acid reflux is a condition shown in millions of people' - Abstract mentions: 'understanding how the phagosome pathway and Cu region influence gene expression may help clarify the genetic regulation of gut homeostasis' in context of acid reflux - Introduction explicitly states: 'our study analyzes gene expression patterns… to better comprehend the molecular mechanisms behind this condition [acid reflux]' - Discussion states: 'Our findings provide valuable insight into the genetic basis of acid reflux' - Acid reflux is a named digestive disorder affecting the gut/metal processing

---

Logan et al. (2025) - CCC FA25.txt

Title: ANCE Gene Expression in Drosophila Midgut

Evidence for this categorization: - Primary focus: ANCE gene (ortholog of human ACE) and cardiovascular disorders - Introduction: 'Cardiovascular disorders are a prominent and widespread disease caused by genetic predispositions and environmental conditions' - Introduction: 'The ACE gene found within humans is an angiotensin converting enzyme, which is a critical regulator of blood pressure and fluid balance' - Abstract: 'studying ANCE gene expression and functional studies may offer insights into cardiovascular-related pathways' - ACE regulates blood pressure - directly linked to cardiovascular disease

---

Otala et al. (2025) - LU SP25.txt

Title: Fly Gut is a good model for fatty acid metabolism disorders

Evidence for this categorization: - Primary focus: Fatty acid metabolism and related disorders - Introduction explicitly states: 'Fatty acid metabolism is essential for energy production, membrane structure, and cellular signaling. Related disorders and symptoms include: MCADD and Carnitine transporter deficiency, hypoglycemia, liver dysfunction, and muscle weakness' - Introduction: 'Dysfunction may lead to fat accumulation, liver disease, and inflammation' - Abstract: 'Fly Gut is a good model for fatty acid metabolism disorders' - The research is explicitly motivated by understanding metabolic disorders

---

Rodriguez (2025) - CCC SP25.txt

Title: Differential Gene Expression of Galm1 in the Drosophila Midgut

Evidence for this categorization: - Primary focus: GALM gene and galactosemia - Abstract: 'Mutations in the protein galactose mutarotase, or GALM, are associated with the disease galactosemia' - Introduction: 'Galactosemia is a rare metabolic disorder that results from deficiencies in the GALM gene' - Discussion: 'Because Galm1 is expressed the most in areas specializing in hexose metabolism… they will not only have a higher risk of developing galactosemia' - Research explicitly motivated by understanding galactosemia

---

Trevino et al. (2022) - CCC SP22.txt

Title: Drosophila Melanogaster a Good Model System of Zellweger Spectrum Disorder

Evidence for this categorization: - Primary focus: Zellweger Spectrum Disorder (ZSD) - Abstract: 'Zellweger Spectrum Disorder (ZSD) is a genetic disorder that is caused by the mutation of 1 out of 13 different genes involved in the formation and function of peroxisomes' - Introduction: 'ZSD is a rare inherited disorder characterized by the absence/reduction of functional peroxisomes in cells' - Discussion mentions peroxisomal disorders and metabolic abnormalities - Research explicitly motivated by understanding ZSD

---

6. Disease research (Other)

Paderna et al. (2025) - CCC SP25.txt

Title: Albinism and Melanogenesis Pathway: Analysis of Differential Gene Expression

Evidence for this categorization: - Primary focus: SLC45A2 gene associated with oculocutaneous albinism type 4 - Introduction: 'In humans, the gene SLC45A2 is associated with oculocutaneous albinism type 4, a condition that affects the skin, eyes, and hair' - Abstract: 'In humans, oculocutaneous albinism type 4 (OCA4) manifests as reduced pigment in the skin, eyes, and hair' - Research explicitly motivated by understanding albinism - Discussion: 'In humans, the gene is related to melanin synthesis (Le et al.). Its ortholog in humans, SLC45A2, is associated with melanin synthesis, and mutations can result in albinism'

Paramo-Ojeda et al. (2025) - CCC SP25.txt

Title: Ras85D Expression in Drosophila Midgut Highlights a Model for KRAS-Driven Colon Cancer

Evidence for this categorization: - Primary focus: KRAS-driven colon cancer - Abstract: 'KRAS mutations drive many human cancers by promoting uncontrolled cell growth' - Introduction: 'One of the most frequently mutated oncogenes in human cancers is KRAS' - Abstract: 'Ras85D… would serve as a great model for gene changes in Drosophila' - Research explicitly motivated by understanding cancer mechanisms - Discussion: 'Further investigation into Ras85D mutations or overactivation in Drosophila could provide insight into the steps of tumor initiation'

Methodology Notes

Category Definitions

Basic biology (Neurology): Posters centered on nervous-system signaling, sleep, feeding behavior, gut-brain communication, or related signaling pathways without a named disease being the main motivating factor.

Basic biology (Metabolism): Posters focused on sugar, lipid, or amino-acid metabolism; digestion; absorption; or energy balance without a named disease being the main motivating factor.

Basic biology (Other): Research focused on understanding fundamental biological mechanisms — such as gene expression patterns, regional tissue specialization, signaling pathways, gene function, patterning, pigmentation, transcriptional control, developmental processes, or cell biology — without explicitly targeting a specific human disease and that do not fall into the basic biology (neurology) or basic biology (metabolism) categories.

Disease research (Neurology): Research motivated by understanding the molecular basis of brain or nervous system disorders. These posters investigate genes and pathways linked to cognitive, behavioral, or neuropsychiatric conditions in humans.

Disease research (Metabolism): Research motivated by understanding metabolic or digestive disorders, including conditions affecting the liver, gut, or nutrient processing. These posters investigate genes and pathways linked to metabolic conditions in humans.

Disease research (Other): Posters where a named human disease, disorder, or direct health model is the main reason for the work, but the poster does not fit into disease research (neurology) or disease research (metabolism).

Other: Posters that do not fit into the categories above.

Analysis Approach

1. All 27 poster texts were read in full before any categorization decisions were made
2. Each poster was assigned to exactly one category based on the primary research motivation
3. When a poster showed elements of multiple categories, the category that best captured the primary motivation was selected
4. The author and reference sections were excluded from analysis as specified
5. Categories were applied consistently across all posters without reference to previously categorized posters

---

Report generated by Minimax M2 AI for educational research analysis